MILITARY SPECIFICATION
CARTRIDGE, CALIBER .30, REFERENCE

1. SCOPE

1.1 This specification covers a cartridge to be used for calibration purposes in Caliber .30 proof testing equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

Munitions Command

C8595416	- Cartridge, Caliber .30, Reference
C7643674	- Classification of Cartridge Case Defects
F10542286	- Packing and Marking, Cartridges Cal. .30 Reference, Cartons, Box Ammunition M2A1, Box Wirebound
IEL-8595416	- Inspection Equipment List for Cartridge, Caliber .30, Reference

PUBLICATIONS

Munitions Command

TECP 700-700, Vol. III	- Manual of Test Methods for Small Arms Ammunition
AMCR 715-505, Vol. 3	- Ammunition Ballistic Acceptance Tests Methods Volume 3, Test Procedures for 7.62MM Cartridges

FSC 1305

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with the requirements specified on Drawing C8595416, referenced specifications and the following:

3.2 Bullet extraction.- The force required to extract the bullet from the cartridge case shall be not less than 45 pounds.

3.3 Residual stress.- The cartridge shall not split when subjected to one percent mercurous nitrate solution for 15 minutes.

3.4 Waterproof.- The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 5 pounds per square inch (psi) for 15 seconds.

3.5 Accuracy.- The average of the mean radii of all targets of the sample cartridges fired at 600 yards shall not exceed 6 inches.

3.6 Action time.- The action time (overall primer ignition, propellant burning and bullet-barrel time) of the cartridge shall not exceed 4 milliseconds.

3.7 Velocity.- The grand average of the velocities of all the sample cartridges, conditioned at 70° ± 2° Fahrenheit (F), shall be 2740 feet per second (ft/sec), plus or minus 20 ft/sec, at 78 feet from the muzzle of the weapon. The average of standard deviations of velocities for fifteen 20-round series shall not exceed 20 ft/sec.

3.8 Chamber pressure.- The average chamber pressure of the sample cartridges, conditioned at 68° to 72°F, shall not exceed 50,000 psi.

3.9 Function and casualty.- The cartridge shall function without casualty.

3.10 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.10.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 Inspection Provisions.

4.2.1 Lot.

4.2.1.1 Submission of Product.- The product shall be submitted in accordance with MIL-STD-105. The cartridge lot shall be restricted to cartridges assembled with cases primed with primers from one lot only, using an established charge of propellant from one lot only. The entire lot of reference cartridges shall be assembled and charged on a single loading machine (see 6.2).

4.2.1.2 Size of lot.- The cartridge lot size shall be as agreed upon between the supplier and the contracting officer and shall not exceed 500,000 cartridges.

4.2.1.3 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity.

4.2.2 Examination.- Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

4.2.2.1 Classification of defects.- The visual and dimensional defects shall be classified as specified in Table I.

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Table I

No.	Defect and Method of Inspection	Major	Minor	Major or Minor
	Visual 1/			
	Cartridge			
1	Discolored, dirty, oily, smeared		X	
2	Corroded; or stained, if etched	X		
	Case			
4	Round head	X		
5	Dent			X
6	Split case	X		
7	Perforated case	X		
8	Draw scratch			X
9	Scratch		X	
10	Beveled underside of head	X		
11	Case mouth not crimped in cannelure	X		
12	Scaly metal			X
13	No chamfer on head (rim)	X		
14	Fold		X	
15	Wrinkle		X	
16	Buckle		X	
17	Bulge		X	
18	Illegible or missing head stamp		X	
19	Defective head		X	
20	Defective mouth		X	
21	No visible evidence of mouth anneal	X		
	Bullet			
22	Dent		X	
23	Scratch		X	
24	Split bullet jacket	X		
25	Loose bullet	X		
26	Missing cannelure	X		
27	Scaly metal			X
28	Upset (crooked) point		X	
29	Exposed steel (clad jacket)		X	
30	Blunt point		X	
31	Defective cannelure		X	

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Table I (Cont'd)

No.	Defect and Method of Inspection	Major	Minor	Major or Minor
	Primer			
32	No primer	X		
33	Cocked primer	X		
34	Inverted primer	X		
35	Loose primer	X		
36	Nicked or dented primer		X	
37	No waterproofing material (primer pocket joint)		X	
38	Defective crimp		X	
	Gaging			
39	Total length	X		
40	Cartridge profile failure (requiring more than 20 lbs dead weight to insert in profile and alignment gage)	X		
41	Diameter of extractor groove, max.	X		
42	Diameter of extractor groove, min.		X	
43	Diameter of head	X		
44	Thickness of head	X		
45	Length to shoulder datum	X		
46	Depth of primer	X		
	Weighing			
47	Weight, under min.	X		

1/ Refer to MIL-STD-636 for Visual Standards for defects 1 through 38. In the event of conflict between Table I of this specification and MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

4.2.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.3.

4.2.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

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Table II

Test	Number of Cartridges	Requirement Paragraph
Bullet extraction 1/	25	3.2
Mercurous nitrate (Residual stress) 1/	20	3.3
Waterproof 2/	50	3.4
Accuracy 3/	90	3.5
Action time 1/	50	3.6
Velocity 4/	300	3.7
Chamber pressure 4/	300	3.8
Function and casualty 5/	-	3.9

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of ten or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than three but less than ten cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample, ten or more cartridges fail to comply with the applicable requirement.

3/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

4/ Failure of the cartridges to comply with the applicable requirements shall be cause for rejection of the lot (see 6.2.1).

5/ Examination for firing defects shall be made during the firing of all ballistic tests. A lot shall be rejected when firing defects exceed the acceptance number in Table III.

4.2.3.2 Firing defects.- Firing defects and acceptance number shall be as specified in Table III.

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Table III

Defects	Acceptance Number
1. Misfire	1
2. Bullet remaining in bore	0
3. Primer leak	
a. Perforation in firing pin indent in primer cup	3
b. Escape of gas through primer cup other than 3a	1
c. Escape of gas around primer cup	
(1) 50% or more than 50% of periphery	10
(2) Less than 50% of periphery	20
d. Blown primer or primer falls out of pocket on retraction of bolt	0
e. Primer remains in pocket, but is physically loose	5
4. Case casualties	
a. Longitudinal split 1/	
(1) Neck and shoulder (I & S)	20
(2) Body (J)	2
(3) Body (K)	1
(4) To head (L)	0
(5) Through head (M)	0
b. Circumferential rupture 1/	
(1) Partial, shoulder or body (J or S)	1
(2) Partial, body (K)	0
(3) Partial, head (L)	0
(4) Complete	0
Total of defects 3c.(1), 3c.(2) and 4a.(1)	30
Total of all other listed defects	5

1/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

4.2.4 Packaging, packing, and marking inspection.- During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrences of a high pressure test cartridge shall be classed as a critical defect. Occurrence of any type other than high pressure test shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing, and marking shall be in accordance with MIL-STD-644 as applicable to the drawing or as required by the contract.

4.2.5 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-8595416.

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4.3 Test methods and procedures.

4.3.1 Bullet extraction.- The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute. The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.3.2 Mercurous nitrate.- Tests shall be conducted in accordance with TECP 700-700, Vol. III.

4.3.3 Waterproof.- The ammunition to be tested, not to exceed five cartridges at one time, shall be placed horizontally on a perforated tray. The tray shall be placed in a desiccator containing sufficient quantity of freshly boiled water to maintain a head of 2 to 2-1/2 inches. The desiccator shall be evacuated to a pressure of 5 pounds per square inch (10 inches of mercury) below atmospheric pressure and held at that pressure for fifteen seconds. The number of bubbles liberated from the mouth or primer of each cartridge shall be observed.

4.3.4 Accuracy.- The test shall be conducted in accordance with TECP 700-700 Vol. III, utilizing the longest range available up to 600 yards. Ranges shorter than 200 yards shall not be used. When accuracy testing is conducted on ranges shorter than 600 yards, the average of the mean radii of the targets shall be multiplied by the applicable conversion factor given below:

Range (yards)	200	300	400	500
Factor	3.40	2.23	1.68	1.35

4.3.5 Action time.- The test shall be conducted in accordance with Publication AMCR 715-505.

4.3.6 Velocity.- The test shall be conducted in accordance with TECP 700-700, Vol. III and the following: The cartridges shall be fired in barrels previously fired 100 to 300 times. Twenty cartridges shall be fired in each of five pressure-barrel assemblies containing the long piston. The average velocity and standard deviation of velocities for each pressure-barrel assembly and the average velocity for the five assemblies shall be computed. If the average velocity for any pressure-barrel assembly varies from the average for the five assemblies by more than plus or minus 14 ft/sec, on the first day of firing, then that assembly shall be discarded and another selected and firing conducted therein until the aforementioned condition is met by all 5 assemblies; thereafter on the 2 succeeding days, the same 5 assemblies shall be used. An additional twenty cartridges shall be fired in the same manner in the same pressure-barrel assemblies on each of two succeeding days. The average velocity and standard deviation of velocities for each twenty round test, the daily average velocity

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(twenty rounds in each of five assemblies) and the grand average for the entire fifteen velocity series (five assemblies on each of three days) shall be computed.

4.3.7 Chamber pressure.- The cartridges shall be drilled, using the specified equipment and the drilled hole shall be covered with a half inch square of tape. The tests shall be conducted in accordance with TECP 700-700, Vol. III and the following: The cartridges shall be fired in barrels previously fired 100 to 300 times. Twenty cartridges shall be fired in each of five chamber pressure barrel assemblies on each of three days. Upon completion, the average pressure for the entire fifteen pressure series (five assemblies on each of three days) shall be computed. The velocity and pressure tests may be fired alternately in the same weapon during the same day.

4.3.8 Function and casualty.- The test shall be conducted simultaneously with all ballistic tests.

4.3.9 Defect penalty.- In any ballistic test in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Level A (Worldwide shipment). The cartridges shall be packed in accordance with Drawing F10542286 or as required by contract.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the drawing cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids or request for proposal and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

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6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

6.2 Uniformity of propellant charge.- A machine shall be chosen which is known to drop charges with better than average uniformity and which gives consistent results in the daily production tests. If relative machine performance is unknown, special tests should be fired to assure selection of a good machine.

6.2.1 A quantity of 20 rounds shall be fired for velocity and 20 rounds for pressure each four hours of an eight hour shift to verify uniformity of charge weight and established velocity and pressure values. Results of these production control firings shall be available upon request. Charge weight, when once established, shall not be changed during production of the entire lot.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1305-A429

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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